

The effect of priming duration on unconscious stimuli recall:

**Assessing the effectiveness of subliminal advertising through the use of the
word-stem completion task**

Group 3

Discussion section 303

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Psych 505: Learning and Decision-Making – Spring 2026

Abstract (Contributed by: Ashrita Patil)

Subliminal advertising relies on the assumption that briefly presented stimuli, undetectable to conscious awareness, can influence subsequent behavior through priming. However, the conditions under which subliminal priming reliably occurs remain unclear, and the threshold separating unconscious from conscious processing is poorly characterized. We tested whether prime duration affects unconscious recall in a within-subjects word-stem completion task, with university students exposed to five-letter prime words at 17, 34, 51, and 1000 milliseconds (ms), followed by self-reported "Guess", "Know", or "Remember" judgments. We hypothesized that the shortest duration (17 ms) would produce the highest rate of unconscious recall, indexed by correct "Guess" responses, while longer durations (34, 51, and 1000 ms) would shift performance toward conscious recognition. A one-way multivariate analysis of variance (MANOVA) showed that conscious recognition ("Remember" and "Know") increased significantly with duration, but unconscious recall ("Guess") did not differ reliably across conditions. These findings suggest that conscious recognition scales with stimulus exposure while behavioral evidence for unconscious recall does not reliably change with priming duration, undermining broader claims about the everyday efficacy of subliminal advertising.

Introduction (Contributed by: Maia-Maria Stanciu)

Subliminal advertising is an important tool for manipulating behavior. Many claims have been made regarding the mechanisms through which subliminal priming operates, and its subsequent effects on the viewer's acts. One example relates to James Vicary, who publicly stated that flashing a message for $1/3000^{\text{th}}$ of a second influenced sales. Vicary claimed to have conducted experiments on subliminal advertising, and one famous example of this is the study conducted at Fort Lee, New Jersey.¹ Within this study, viewers of a film at a movie theatre were primed every 5 seconds, through the use of a tachistoscope connected to a projector.¹ Messages such as "Drink Coca-Cola" and "Hungry? Eat Popcorn" were flashed for $1/125^{\text{th}}$ of a second, presumably resulting in a "18.1% increase in Coca-Cola sales and a 57.8% increase in popcorn sales".¹ Future studies have deemed Vicary's claims a hoax, suggesting that although the subliminal

messaging increased beverage sales, it did not promote preference for Coca-Cola over other alternative brands.² However, findings demonstrate that subliminal advertising can be effective under the right conditions.² One factor found to reliably affect consumer preference via subliminal messaging is goal-directed preference for the shown stimulus.² For instance, thirsty individuals were more likely to consume a Kool-Aid beverage after being exposed to thirst-related words for 16 milliseconds, such as “thirst” and “dry”, as compared to participants who were not thirsty.³ This finding shows that the efficacy of subliminal advertising is dependent upon specific criteria. Therefore, it is essential to determine the conditions under which subliminal messaging is effective.

Priming stands at the core of subliminal advertising. This can be defined as the unconscious recall of a previously observed stimulus.⁴ Previous research has shown that, within most trials, longer stimuli presentations have a negative effect on priming, such that the duration of stimuli exposure is negatively correlated with priming effects.⁴ The strength of a priming effect can usually be measured through an assessment of the participants’ reaction time and accuracy as it relates to the stimulus of interest.⁴ Moreover, shorter stimuli presentation can increase the accuracy of implicit recognition in later assessment. When displaying images for 250 or 350 milliseconds, researchers observe a significant difference in the amount of ‘guessed’ correct responses during recall, such that the image displayed for a shorter time resulted in a higher number of correctly ‘guessed’ responses.⁴ Within this study the proportion of ‘guessed’, ‘remembered’, and ‘known’ responses, as coded by the participants, determines the conscious awareness of the primed individual.⁴ Therefore, this measure can be implemented to examine whether priming affects unconscious or conscious recall for the previously observed stimulus.

The word-stem completion task is usually implemented for testing the effects of priming in humans.⁵ This task involves providing individuals with a list of word stems, which they are asked to complete by filling in the ending to each word stems with the first letters that come to mind.⁵ Individuals previously exposed to certain words via priming are more likely to fill in the word stems on the basis of those words. For example, individuals primed with the word “Motel” are more likely to fill in the word stem “Mot_ _”

to form the word “Motel”, as opposed to other English common words that they have not been previously primed with, such as “Motor”.⁵

Our experiment will seek to examine the ways in which the duration of the prime words shown to individuals influences the conscious and unconscious recall rates for these specific words, analyzed through the use of the word-stem completion task. Its experimental design will focus on assessing whether a threshold exists at which priming duration undermines the effectiveness of subliminal advertisement viewing in a university student population. Determining this threshold will be done by exposing participants to three experimental conditions. Specifically, the durations of the prime words will be 17, 34, 51, and 1000 milliseconds (ms). Subliminal effects have been found at 17 ms, as demonstrated in a study conducted by Ruch et. al.⁶ These experimental conditions were chosen as each represents one, two, or three frames on a 60 Hertz (Hz) display respectively. Exposure to the prime words for 1000 ms will be implemented as the control. Unconscious and conscious recognition rates will be measured by performance on the word stem completion task and the participants’ answer on a survey that asks participants whether they guessed, knew, or remembered the words they formed on the abovementioned task.

We hypothesize that participants exposed to a shorter stimulus duration (17 ms) will show a higher unconscious recognition rate for that stimulus, as opposed to participants exposed to a longer stimulus duration (34 or 51 ms) or the control condition (1000 ms). Moreover, participants moderately exposed to the prime (34 ms) will display an unconscious recognition rate lower than that of participants in the shorter stimuli duration condition (17 ms) but higher than that of participants in the longer stimuli duration condition (51 ms), or the control condition (1000 ms). Participants exposed to a shorter stimulus duration (17 ms) will show a lower conscious recognition rate for that stimulus, as opposed to participants exposed to a longer stimulus duration (34 or 51 ms) or the control condition (1000 ms). Finally, participants moderately exposed to the prime (34 ms) will display a conscious recognition rate higher than that of participants in the shorter stimuli duration condition (17 ms) but lower than that of participants in the longer stimuli duration condition, (51 ms) or the control condition (1000 ms).

Results (Contributed by: Maia-Maria Stanciu, Ashrita Patil)

Fig. 1: "Guess" accuracy was numerically highest at the shortest prime duration.

Duration	Guess (Unconscious)
17ms	50.0% (6/12)
34ms	40.0% (8/20)
51ms	31.3% (5/16)
1000ms	33.3% (4/12)

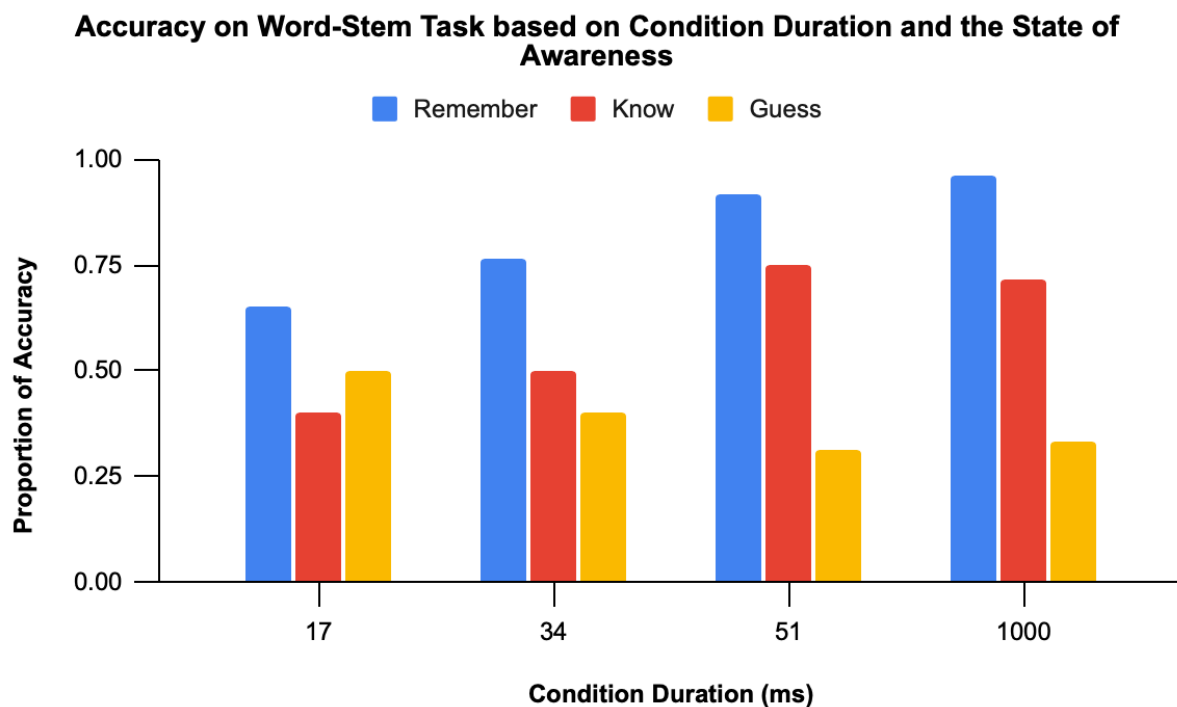
The number of correctly guessed prime words on the word-stem completion task is plotted by duration condition (17, 34, 51, and 1000 ms). Accuracy peaked in the 17 ms condition and was lowest in the 51 ms condition, a descriptive pattern consistent with subliminal priming at the shortest duration.

Fig. 2: "Remember" accuracy increased monotonically with prime duration.

Duration	Remember (Conscious)
17ms	65.2% (15/23)
34ms	76.5% (13/17)
51ms	92.0% (23/25)
1000ms	96.2% (25/26)

The number of correctly remembered prime words is plotted by duration condition. Accuracy was highest in the 1000 ms control condition and lowest in the 17 ms condition, consistent with conscious memory scaling with stimulus exposure.

Fig. 3: Response category accuracy across all four duration conditions.



Correctly identified prime words are plotted by duration for each of the three self-report categories ("Guess", "Know", and "Remember"). "Remember" accuracy increased approximately linearly with duration, "Know" accuracy increased and plateaued near 51 ms, and "Guess" accuracy declined from 17 to 51 ms before rising slightly at 1000 ms.

Fig. 4: One-way MANOVA results for the effect of prime duration on response category accuracy.

Subjective State	df	F	p	η^2 ▲	Interpretation
Guess	(3, 36)	0.41	0.746	0.03	Non-Significant; time-independent
Know	(3, 36)	4.12	.018*	0.26	Significant increase with early plateau
Remember	(3, 36)	12.44	< .001*	0.51	Significant linear increase; time-dependent

Duration significantly affected "Remember" accuracy ($p < 0.001$) and "Know" accuracy ($p = 0.018$), but did not significantly affect "Guess" accuracy ($p = 0.746$).

The number of correctly guessed prime words was numerically highest in the 17 ms condition and lowest in the 51 ms condition (Fig. 1). The number of correctly remembered prime words was highest in the 1000 ms condition and lowest in the 17 ms condition (Fig. 2). Across all three response categories, "Remember" accuracy increased approximately linearly with duration, "Know" accuracy increased and plateaued near 51 ms, and "Guess" accuracy declined from 17 to 51 ms before rising slightly at 1000 ms (Fig. 3).

A one-way MANOVA tested the effect of prime duration on each response category (Fig. 4). Duration significantly affected "Remember" accuracy ($p < 0.001$) and "Know" accuracy ($p = 0.018$), but did not significantly affect "Guess" accuracy ($p = 0.746$).

Discussion (Contributed by: Brennen Haynes, Ashrita Patil)

We used the word-stem completion task to test whether subliminal exposure to prime words produces measurable priming effects, and whether this effect varies with stimulus duration. "Remember" and "Know" responses index explicit recognition, while "Guess" responses signify the absence of retrieval awareness, or unconscious recall.⁷ Using a within-subjects design, we found that conscious recognition scaled with duration: participants correctly remembered significantly more prime words at longer durations, with the strongest performance in the 1000 ms control condition.

Our hypothesis predicted three patterns across the duration conditions. First, "Remember" rates were predicted to increase with duration, with the highest accuracy in the 1000 ms control condition. This prediction was supported: "Remember" accuracy scaled significantly with duration ($p < 0.001$), with the strongest performance at 1000 ms, confirming that conscious memory tracks stimulus exposure. Second, "Know" rates were predicted to increase with duration, as this also signifies conscious recall. This prediction was partially supported: "Know" accuracy did increase significantly with duration ($p = 0.018$), but plateaued near 51 ms rather than rising monotonically. Third, and most centrally, the shortest duration (17 ms) was predicted to produce the highest rate of unconscious recall, indexed by correct "Guess" responses. This prediction was not statistically supported. Although "Guess" rates were numerically highest at 17 ms,

the MANOVA returned a non-significant effect of duration on "Guess" accuracy ($p = 0.746$); the slight uptick at 1000 ms is most likely noise.

A natural extension is to pair the word-stem completion task with simultaneous electroencephalography (EEG). EEG records electrical activity directly from the scalp on a millisecond timescale, which aligns with the timing of our duration manipulation in a way that hemodynamic methods like fMRI cannot.⁸ More importantly, the remember/know/guess paradigm produces dissociable patterns of neural activity for explicit and implicit recognition. Within the dual-process framework, explicit recognition is indexed by two positive-going components: the mid-frontal old/new effect (also termed the FN400), linked to familiarity, and the parietal old/new effect, linked to recollection.⁹ Voss and Paller further report that highly accurate "Guess" responses, interpreted as evidence of implicit recognition, are associated with an early, negative-going ERP component over occipital sites, dissociable from the positive-going correlates of explicit memory.⁷ Concurrent EEG would generate three falsifiable predictions: correctly guessed items at 17 ms should elicit the negative-going component without a mid-frontal or parietal old/new effect, providing a neural signature of unconscious priming independent of subjective report; correctly remembered items at 1000 ms should elicit the canonical mid-frontal and parietal old/new effects; and the ambiguous "Know" category would resolve into one of two patterns, with mid-frontal-dominant responses indicating weak conscious familiarity, or negative-ERP-dominant responses indicating implicit recognition mislabeled as familiarity. fMRI could provide complementary spatial evidence by leveraging known dissociations within the medial temporal lobe, where the hippocampus is preferentially engaged during recollection and the perirhinal cortex tracks familiarity-based recognition.¹⁰

Several limitations qualify these results. The sample was small and drawn from a single university population, and web-based delivery meant we could not verify that participants were fully attentive to the experimental task. Although we aimed to collect data from around 30 student participants, we could only achieve a sample of 9 individuals due to time constraints, which lowered the generalizability of our study.

Potential confounding variables to our study's results included participants' gender and fluency in the English language. Participants who lack fluency in the English language might find it more difficult to fill in the word-stem task, lowering the internal validity of our study. Moreover, as we aimed for confidentiality and did not collect any participant identification data apart from the randomly generated participant ID or their initials, we could not analyse gender as a potential variable that influenced our findings. The within-subjects design may have introduced practice effects, as each individual was instructed to complete one condition duration at a time before advancing to the next one, and this could have made participants more perceptive of the latter prime words presented. Future behavioral work could address these limitations with a larger and more diverse sample, the presentation of a questionnaire inquiring about gender and English fluency before the start of the task, and a between-subjects design. Selecting participants through convenience sampling possibly introduced selection effects and researcher bias. Therefore, conducting future studies through random sampling can help solve this problem.

Collectively, our findings provide strong evidence for conscious recall scaling with prime duration but only weak and statistically non-significant evidence for the impact of priming duration on unconscious priming. This is broadly consistent with the literature suggesting that subliminal advertising is effective only under specific, narrowly defined conditions^{2,3} and offers little support for the broad public concern that subliminal messaging meaningfully manipulates behavior in everyday viewing contexts.

References (Contributed by: Maia-Maria Stanciu, Brennen Haynes, Ashrita Patil)

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Methods (Contributed by: Maia-Maria Stanciu, Brennen Haynes)

In our task, we tested the effectiveness of subliminal messaging and its ability of priming participants for the completion of the word-stem task. Our participants were selected from the population by non-probability sampling, specifically convenience sampling, due to time constraints. An email was individually sent to a group of UW-Madison university students to request their participation in our study, which included a public link to our experiment. Individuals were asked to participate in a study on memory and recall, so as to provide informed consent, but the specific aim of the study had not been revealed to minimize demand characteristics. We aimed to sample about 30 college students, both male and female, as our participants. However, we were only able to achieve a sample of 9 individuals due to time constraints. We used a within-subjects, repeated measure design where all participants completed each condition of the task. The experimental conditions of our task showed a prime word for 17, 34, 51 ms, while the control condition consisted of displaying the prime word for 1000 ms. The order of exposure to the study's conditions was counterbalanced to reduce order effects. The task started with an ethics statement, which provided informed consent of the task, stated that participants can withdraw at any time, highlighted that participation is entirely voluntary, and emphasized the anonymity of the data collected. If participants agreed with the aforementioned terms, they could click an "I agree, proceed" button in order to move onto the next stage of the experiment. Next, participants were instructed to enter a "Participant ID" consisting of their initials or leave the field blank for complete anonymity. The sample was then instructed to fixate their eyes on the "+" at the center of their screen and try not to blink as a group of words would "flash very briefly" before their eyes. Individuals then received five words on the screen for either 17, 34, 51 or 1000 ms, randomly assigned. The words shown within each condition were all common, English, five-letter words. All words were made up of an equal number of letters to reduce the potential risk of word length being a confounding variable in our experimental study. Moreover, the words shown for each participant in each condition were randomly extracted from a word bank consisting of 467 words, in order to reduce the probability of participants being exposed to the same word twice, which could

introduce an internal validity threat. We implemented backward masking after each word, defined as a visual prime being accompanied by a mask shortly after its showing, leading to “the complete absence of the prime’s visibility”.¹¹ Masking was introduced for “attention shifting” from the previous primed word, in order to reduce the probability of carryover effects throughout all experimental conditions.¹¹ After each duration condition, participants were asked to fill in ten word stems, five of these corresponding to the primed words and five being random word stems uncorrelated with the primed words shown. For each word stem, participants were asked to insert exactly two letters at the end to form an English word.

Next, we asked participants via a close-ended questionnaire whether their completion of each word from the word stem task was because they held a vague familiarity of the word, corresponding to “Know”, explicitly remembered vivid details of seeing the word, corresponding to “Remember”, or completely guessed how to complete the word stems, corresponding to “Guess”. As the experiment held a within-subjects design, each participant went through every duration condition and was asked to complete both the word-stem task and the questionnaire after each of the four total conditions.

Data was collected and analyzed for accuracy, as defined by how many word stems demonstrated a priming effect. Therefore, our independent variable was the duration condition, its four levels being 17 ms, 34 ms, 51 ms, and 1000 ms, while our dependent variables were the priming success, coded as participants either completing the word-stem task in accordance with the previously shown prime or failing to do so, and the survey answers, which could be either “Guess”, “Know”, or “Remember”. A one-way multivariate analysis of variance (MANOVA) was conducted to establish whether there was a significant relationship between the word-stems correctly completed in accordance with the prime words shown, the priming duration condition, and the participants’ survey answers of “Guess”, “Know”, and “Remember”.

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